

Start

a few offers per device: its baseline plan,
and its price responses at the import and
export prices

One iteration, repeated

Coordinator LP

meter: import, export, limits, curtailment;
LP devices (the battery) as exact variables
keeps every offer and picks the *blend* with
the cheapest bill plus private costs

prices π , one per slot

offers: plan + private cost

Each device that bids (tank, HVAC, EV)

its price response: its best plan if a kWh in
slot t costs π_t , by any solver
it never sees the tariff or the other devices

their answers give a lower
bound; stop when blend -
bound is small

Result

one runnable plan per device (a blend,
where the device can run one), then a
polish
*certificate: at most the gap from the best
possible*